From Scores to Kinds: Projectibility and Validity in AI Evaluation

Brett Reynolds [*]
Humber Polytechnic & University of Toronto

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Abstract

AI evaluation terms often let scores and labels travel beyond the cases that produced them. A benchmark result becomes evidence for REASONING; an attack result becomes evidence of JAILBREAK RESISTANCE; an aggregate label becomes a governance claim about ALIGNMENT or TRUSTWORTHINESS. This paper argues that such terms function as inference licences. They authorize movement from observations, outputs, interactions, or scores to claims about recurrence, vulnerability, capability, or governance.

Combining construct validity with Boyd's and Khalidi's accounts of projectible kinds, I ask what each term licenses and what would let that licence travel across models, benchmarks, attacks, and deployments. I develop a diagnostic typology: output-centred failure profiles, adversarial relations, capability attributions, and umbrella governance constructs. The typology clarifies why disputes about HALLUCINATION, JAILBREAK, REASONING, EMERGENCE, AGENCY, ALIGNMENT, and TRUSTWORTHINESS often go wrong: one inferential profile is read as another. The payoff is a way to design and report AI evaluations around the inferences they can support.

1 Introduction

AI evaluation often moves from a score to a much larger claim. A benchmark result is reported as evidence for REASONING. A safety evaluation says that a system has JAILBREAK RESISTANCE. A model card, audit, or policy document describes an assistant in terms of ALIGNMENT, TRUSTWORTHINESS, or HALLUCINATION risk. These labels point to observed behaviour and invite readers to draw inferences: that a failure will recur, that a defence will generalize, that a capability will transfer, or that a system is suitable for some governance purpose.

The debate over emergent abilities in large language models shows the problem in a compact form. Wei et al. (2022) argue that some abilities appear only once models reach sufficient scale. Schaeffer et al. (2023) reply that some apparent jumps can be produced by the choice of metric: discontinuous scoring can make gradual changes in performance look like sudden emergence.

[*]Contact: brett.reynolds@humber.ca

The dispute reaches beyond the acquisition of new abilities. It asks what a score licenses us to infer. If a benchmark curve has a sharp bend, can we infer a new capability, a change in the measurement instrument, or something more local about a task, metric, and model family?

That example is one instance of a broader pattern. Raji et al. (2021) argue that influential benchmarks are often treated as stand-ins for broad claims about general ability, even though the benchmarks themselves are finite, contextual, and constructed around specific tasks. A benchmark can be useful while still being too narrow to support the claim attached to it. AI evaluation terms are vague in places, but the deeper issue is inferential. They function as inference licences: they authorize a move from an observation or score to a claim about recurrence, vulnerability, capability, or governance. Different terms authorize different moves.

This paper asks which inference each AI evaluation term licenses, and what would make that inference projectible. I use projectibility in a plain sense first: a classification is projectible when applying it in one case gives us a warranted reason to expect something in another.

If a system produces one unsupported answer, that alone doesn't show a general tendency to hallucinate. If an attack succeeds once, that alone doesn't show that the model is generally unsafe. If a model passes one reasoning benchmark, that alone doesn't show a stable reasoning capability. The classification becomes useful when we can say what else should follow from it, in which domain, and under which limits.

The philosophical background sharpens this point. In a classical definition, a term gives necessary and sufficient conditions for membership. Boyd's account of homeostatic property clusters and Khalidi's causal-network account show how classes can support projection without that form of definition (Boyd, 1991, 1999; Khalidi, 2018). For AI evaluation, the lesson is modest: useful terms needn't have essences, but they should mark structures that support defensible inferences beyond the immediate observation.

The construct validity tradition gives the measurement side of the same problem. A test score doesn't interpret itself. It needs a construct, an operationalization, a metric, a population of instances, and an intended use. Kane's argument-based view of validation is useful here because it treats validity as a question about the interpretations and uses licensed by scores (Kane, 2013). AI evaluations should be read the same way. A label such as hallucination or jailbreak belongs to a measurement practice that includes tasks, prompts, raters or automated judges, policies, metrics, populations, and deployment decisions.

The division of labour is the paper's hinge. Construct validity asks what a score or label licenses in the measured setting. Projectibility asks when that licensed inference can travel to another model, benchmark, attack, or deployment. The first question concerns the local interpretation of a result. The second concerns the conditions under which that interpretation can support a further expectation elsewhere.

The paper develops a four-way typology of AI evaluation terms by their primary inferential role: output-centred failure profiles, adversarial relation kinds, capability-attribution terms, and umbrella governance constructs. The typology is deliberately non-exclusive. It classifies a term by the inference its evaluations are mainly used to license, not by a metaphysical essence or by all of the term's secondary associations.

The argument proceeds as follows. Section 2 presents evaluation terms as inference licences and connects them to construct validity. Section 3 introduces projectibility without essences, using Boyd and Khalidi to explain why some classes can support inference without classical definitions. Section 4 states the typology. Sections 5–7 work through hallucination, jailbreak resistance, and capability attribution. Section 8 uses sycophancy as a pressure case for alignment and trustworthiness as umbrella constructs. Section 9 returns to the central diagnostic: an AI evaluation term is useful when we know what it licenses, what makes that licence travel, and where it stops.

2 Evaluation terms as inference licences

Evaluation language climbs an inference ladder. A raw output, interaction, or score becomes an evaluation label. The label receives a construct interpretation. The construct supports a kind attribution. The attribution then enters a model comparison, release decision, procurement standard, or governance claim. Each step can be legitimate, but the warrant changes as the claim climbs.

raw output, interaction, or score
↓
evaluation label
↓
construct interpretation
↓
kind attribution
↓
comparison, release, procurement, or governance claim

Figure 1: The inference ladder in AI evaluation.

Consider a mathematical-reasoning benchmark. Raw completions are first coded as right or wrong on individual items. The metric aggregates those judgements into an accuracy score. The benchmark report then interprets the score as evidence for a construct such as MATHEMATICAL REASONING. A leaderboard turns that interpretation into a model ranking, and a user can then treat the ranking as evidence for deployment suitability. The same number appears throughout, but the claim has changed at every rung.

The narrow inference can be sound while the wider one fails. A high score on one population of word problems can license a claim about performance on those items, under that prompt format and scoring rule. It doesn't yet license a general reasoning claim across paraphrases, irrelevant clauses, new item generators, languages, tool scaffolds, or contamination controls. The ladder therefore makes the analytic question visible: which movement is being licensed, and what evidence supports it?

Construct validity names the first warrant. A score has meaning only relative to a construct, an operationalization, a metric, a population of instances, and an intended use (Chouldechova et al., 2024; Kane, 2013). An operationalization is the concrete test procedure that makes a construct measurable. Here, validity is local: the relevant warrant connects a score to a stated interpretation and use. Work on LLM benchmarks makes the same point in the evaluation setting: a benchmark operationalizes a phenomenon through tasks and metrics, then supports a claim only if those parts line up (Bean et al., 2025; Freiesleben, 2026).

In LLM evaluation, repeated probing often makes RELIABILITY easier to estimate: evaluators can ask whether results are stable across item samples, prompt variants, decoding settings, and evaluator systems. Reliability sharpens the local warrant, but it doesn't settle interpretation. The validity question is which probes are compressed into which score, for which construct interpretation and use.

Model comparison makes that compression consequential. Ranking model A above model B requires more than running the same benchmark twice. It requires a judgement that the chosen score summarizes the right profile of tasks, prompts, languages, evaluator systems, and deployment conditions for the intended comparison. A leaderboard can be valid as a narrow benchmark comparison and fail as a governance conclusion if those supports don't hold (Raji et al., 2021).

The comparison point also marks the limit of construct validity. Validity can say what the score licenses here, but it doesn't settle whether the claim travels to another model family, attack distribution, user population, or deployment. That's the projectibility question.

3 Projectibility without essences

The projectibility question turns on structure rather than classical definitions. A classical definition gives necessary and sufficient conditions for class membership. The accounts I draw on reject that requirement. Boyd's account of homeostatic property clusters explains how a class can support projection when its properties cluster through stabilizing mechanisms. Khalidi's causal-network account explains projectibility through relations among properties, where some properties are more central and help explain the presence or stability of others (Boyd, 1991, 1999; Khalidi, 2018).

For this paper, the shared lesson is that an evaluation term can be useful without a crisp definition if it marks structure that supports reliable inference.

Here, class and kind do different jobs. A class is any grouping introduced by an evaluation practice. A kind, in the stronger sense used here, is a class whose structure supports projection. The paper therefore doesn't ask whether every label is a kind. It asks when an evaluation class earns that stronger status.

A classification is projectible when applying it in one setting gives a warranted reason to expect something in another setting, under stated conditions. The conditions matter. Projection is field-relative: what travels depends on the domain, purpose, population, and use of the evaluation. A label can project across nearby prompt variants while failing to project across languages, tool access, deployment settings, or adversary classes.

Suppose a red-team test finds that an adversarial suffix (a string appended to a prompt) makes a chat assistant answer a prohibited request. The result supports a narrow projection. If the suffix still works after a few tokens are swapped, after a harmless phrase is added before it, or when another optimizer seed produces a similar appended string, the test gives reason to expect similar vulnerability under the same chat interface, refusal policy, and evaluator standard.

The same result says little about an indirect prompt-injection case where a browser agent reads a web page that tells it to send a private file. The attacker, channel, policy boundary, and system boundary have changed, so the original classification no longer carries the same expectation.

In AI evaluation, projectibility-relevant structure is artificial, engineered, and institutional. It can include training data, model architecture, training objectives, post-training incentives, decoding procedures, retrieval design, prompt format, interface constraints, policy boundaries, evaluator models, benchmark construction, reporting conventions, and deployment use. None of these is necessary or sufficient; none is essential. Together, though, they can explain why a failure profile, vulnerability, capability attribution, or governance bundle recurs, and why projection weakens when those supports change.

A structure matters for projectibility when it does two jobs: it explains why an inference travels and where it stops. A resampling detector gives the simple case. The detector treats variation across sampled answers as evidence that a claim is unsupported. The signal can travel to settings where unsupported claims vary across samples. It travels poorly to a model that repeats the same unsupported answer every time, because the signal has disappeared.

The same test applies to jailbreaks and reasoning scores. A jailbreak attack transfers when systems share the interface and refusal machinery that the attack exploits; change the interface, policy filter, or model family, and the basis for transfer changes. A reasoning score travels when item variants test the same capacity; if irrelevant clauses or a different metric reverse the result, the score tracked the test setup more than the capability.

The typology below applies that test to four recurring roles in AI evaluation language. For each role, the question is what inference the term licenses, what structure lets that inference travel, and what change breaks the projection.

4 A typology of AI evaluation terms

The typology classifies AI evaluation terms by their primary inferential role. The relevant question is practical and semantic at once: when evaluators use the term, what are they trying to infer from the observation, score, or label? Table 1 gives the diagnostic form. Each row has the same shape: observation, operationalization, licensed inference, projectibility payoff, characteristic invalid inference, and use context. Read each row from left to right: an observation becomes an operationalization, the operationalization licenses an inference, and the final columns state what can travel and what overgeneralization should be blocked.

Type	Observation	Operationalization	Licensed inference	Projectibility payoff	Invalid inference	Use context
Output-centred failure profile	generated output	factuality, faithfulness, support, uncertainty, granularity	the failure is likely to recur under specified structural supports	predicts recurrence, detection, and mitigation transfer under similar generation or grounding conditions	treating HALLUCINATION as false output, or treating one low score as deployment safety	model evaluation, retrieval-augmented generation, factuality claims
Adversarial relation kind	boundary-crossing interaction	attack distribution, policy class, evaluator, target system	the system is vulnerable under a specified adversarial network	predicts vulnerability and defence brittleness relative to attacker class, interface, and policy boundary	treating JAILBREAK RESISTANCE as a model property or projecting across attacker classes	red-teaming, release gates, security evaluation
Capability-attribution term	task performance	benchmark, metric, item population, prompting scaffold	the performance supports a capability claim over a domain	predicts transfer or robustness across item variants, metrics, scaffolds, and system boundaries when the attribution is validated	treating a score as a capability, or a benchmark as the domain	benchmark reporting, model comparison, deployment claims
Umbrella governance construct	aggregate risk or value label	component taxonomy, governance use	lower-level constructs are being coordinated for a purpose	predicts which lower-level evaluations should travel together for a stated governance decision	treating ALIGNMENT or TRUSTWORTHINESS as a single behavioural construct with a single score	procurement, audit, release and risk-management decisions

Table 1: AI evaluation terms by primary inferential role.

The first row covers output-centred failure profiles. HALLUCINATION is the main case: a generated output is assessed for factuality, faithfulness, support, uncertainty, and grounding. The licensed inference concerns recurrence and mitigation transfer under similar generation and grounding conditions. The misanalysis treats hallucination as a bare false-output label or projects from a single benchmark score to deployment safety.

The second row covers adversarial relation kinds. JAILBREAK evaluations describe a relation among a model, interface, policy boundary, attacker, and evaluator. The licensed inference concerns vulnerability under a specified adversarial network. The misanalysis treats resistance to one attack class as general resistance.

The third row covers capability attributions. REASONING, EMERGENCE, and AGENCY move from task performance to claims about what a system can do across a domain. The licensed inference concerns transfer or robustness across item variants, metrics, scaffolds, and system boundaries. The misanalysis treats a benchmark as the domain itself.

The fourth row covers umbrella governance constructs. ALIGNMENT and TRUSTWORTHINESS coordinate lower-level evaluations for institutional purposes. Their projectibility is coordinative: the terms license expectations about which lower-level constructs should be evaluated together for a decision. The misanalysis treats an umbrella label as a single behavioural kind.

The typology is non-exclusive. Hallucination has relational features because unsupportedness depends on a source, a task, and a user context. Jailbreaks often end in bad outputs. Agency can be described as a capability, a system architecture, or a deployment relation. The classification rule is therefore about primary inferential role: a term's type is fixed by the inference its evaluations are mainly used to license.

Confused debates arise when one profile is mistaken for another. A score is read as a capability, a relational vulnerability is read as a model property, or an umbrella governance label is read as a single behavioural cluster. The worked cases make that diagnostic concrete.

5 Hallucination as an output-centred failure profile

Hallucination starts from a generated output, but the failure profile is wider than falsity. It includes factuality, faithfulness to source, support and grounding, uncertainty, granularity, retrieval conditions, task context, and user uptake. The output remains the centre of the evaluation, but the source, retrieval system, prompt, and use context help determine which failure profile the output instantiates (L. Huang et al., 2025; Ji et al., 2023).

The source relation is central. In source-conditioned generation, an output can contradict the source, add claims that the source doesn't verify, or introduce content that's true but unsupported by the assigned evidence. In open-domain question answering, the relevant source can be world knowledge rather than an input document. In retrieval-augmented generation, it can be the retrieved context, the retrieval corpus, or a cited passage. These settings share a family resemblance, but they don't license the same inference.

False output is therefore too coarse. A model can make a factual error, overstate what the available source warrants, contradict a retrieved document, fail to retrieve the needed evidence, answer with misplaced confidence under uncertainty, or give an answer whose harm depends on the user's task. These are related failures, but their operationalizations differ. A factuality metric for biographical claims doesn't measure source faithfulness in summarization, and a source-faithfulness score in retrieval-augmented generation doesn't measure whether a medical or legal assistant should have abstained.

Task goal matters too. If a system is asked to write a true-or-false test and includes "Toronto is the capital of Canada" as an item, the false proposition is intentional test content, not a hallucination. The same sentence becomes a hallucination if the system presents it as true, treats it as supported, or uses it as an answer where accuracy is the goal. The label tracks the role the output is meant to play, not falsity in the string alone.

Hallucination is best treated as a failure profile. The profile has recurring parts: a plausible generated answer, an evidential or factual defect, some relation to a source or task goal, and a user-facing risk created by the answer's uptake. Different evaluations put weight on different parts of the profile. A biography generator invites atomic factual checks; a summarizer invites source-faithfulness checks; a high-stakes assistant invites abstention and uncertainty checks.

The measurement literature already reflects this plurality. FActScore decomposes long-form answers into atomic facts and asks whether those facts are supported by a specified source (Min et al., 2023). SelfCheckGPT uses instability across sampled responses as a black-box signal for unsupported claims (Manakul et al., 2023). Retrieval augmentation changes the failure profile by changing the evidence available to the model, but retrieval can also introduce noisy, missing, or conflicting context (L. Huang et al., 2025; Shuster et al., 2021). Uncertainty-based work gives another signal: under some conditional-generation settings, higher predictive uncertainty tracks greater hallucination risk (Xiao & Wang, 2021).

These operationalizations license different projections. An atomic-fact score can support claims about factual precision under a chosen source. A resampling detector can support claims about unstable unsupported content. A retrieval intervention can support claims about settings where missing parametric knowledge is the relevant bottleneck. Each projection is useful, and each has a failure boundary.

The failure boundary is part of the construct. Resampling can miss a model that consistently repeats the same false claim. Atomic-fact scoring can miss relevance, recall, and whether the system should have answered at all. Retrieval can reduce parametric-knowledge failures while introducing wrong, noisy, missing, or conflicting evidence. These aren't peripheral complications. They specify the conditions under which the hallucination label supports projection.

Classifying hallucination as an output-centred failure profile licenses projection to recurrence under specified generation and grounding conditions. If the same knowledge boundary, decoding procedure, retrieval design, incentive structure, and uncertainty-handling processes are in place, the classification can support expectations about where similar failures will recur and which detection or mitigation techniques can transfer. It blocks the inference from one low hallucination score to deployment safety.

6 Jailbreak as an adversarial relation

Jailbreak starts from an interaction that crosses a policy boundary. The evaluation object is a relation among model behaviour, attacker strategy, interface affordances, guardrail design, evaluator judgement, and the institutional definition of prohibited output. That's why jailbreak resistance should be paired with an adversary model rather than treated as an intrinsic property of a model (Vassilev et al., 2024; Yi et al., 2024).

The relevant relation has several components. There's an attacker goal, an attacker's knowledge of the system, an attacker capability, an interaction channel, a policy boundary, a guardrail or refusal design, an evaluator standard, and a target system. Direct chat jailbreaks, transferable adversarial suffixes, persuasion-based attacks, and indirect prompt injection share enough of this profile to be grouped together, but they don't share one mechanism (Greshake et al., 2023; Zeng et al., 2024; Zou et al., 2023).

This relational structure explains why in-the-wild jailbreaks are a social and technical population. Prompt families are copied, revised, hosted, tested, and patched across time. Their success depends on vendor policy, model update, interface, forbidden-content class, and evaluator choice (Shen et al., 2024). A benchmark that reports attack success rate without this network can still be useful, but the reported number has a narrow inferential target.

Classifying jailbreak as an adversarial relation licenses projection only relative to a specified adversarial network. A result can support expectations about vulnerability under attacker class A, interface I, policy boundary P, and attack distribution D. It doesn't by itself support expectations about attacker class B, a different interface, or a new policy regime. The payoff is a more precise release claim: resistance is always resistance to a defined attack space.

7 Reasoning, emergence, and agency as capability attributions

Reasoning, emergence, and agency share a score-to-capability problem. Each begins with task performance and asks whether that performance supports a claim about what the system can do across a domain. The failure routes differ. Reasoning claims can be fragile across item variants, distractors, scaffolds, and contamination controls. Emergence claims can depend on metric choice. Agency claims can depend on where the model-tool-environment boundary is drawn.

The common risk is treating the benchmark as if it directly revealed the capability. A task score is an observation, a benchmark aggregate is an operationalization, and a capability attribution is an interpretation. The attribution becomes projectible only when the benchmark is embedded in a wider network of evidence: item variation, tests that separate the target construct from nearby constructs, contamination checks, prompting conditions, and comparisons with neighbouring constructs (Freiesleben, 2026).

Reasoning should be the central case. The survey literature treats reasoning as a plural and unsettled construct, covering deductive, inductive, abductive, analogical, causal, probabilistic, formal, and informal reasoning (J. Huang & Chang, 2023). That plurality matters for validity. A benchmark can show strong performance on one task population while leaving open whether the model has a reasoning capacity that transfers across problem forms, domains, and prompting regimes.

The plurality also changes what counts as confirming evidence. A deductive reasoning claim asks for validity-preserving steps across form changes. A causal reasoning claim asks for sensitivity to interventions and counterfactual structure. A mathematical word-problem claim asks for stable mapping from the story to the relevant operations. Treating these as one flat reasoning score makes the construct easier to report and harder to validate.

GSM-Symbolic makes the worry concrete. Mirzadeh et al. (2025) generate controlled variants of GSM8K-style problems and show that model performance can vary substantially across value substitutions, problem complexity, and seemingly irrelevant clauses. These results leave the general reasoning question open while showing that a fixed benchmark score can hide a distribution of behaviours. A reasoning attribution becomes stronger when it survives controlled item variation, contamination checks, distractors, and changes in scaffold.

This fragility matters because the benchmark score is often asked to carry a larger claim than the item population supports. If irrelevant clauses change the answer strategy, the score no longer cleanly measures mathematical reasoning over the intended domain. If chain-of-thought prompting supplies a scaffold, the capability claim should specify whether the target is unaided problem solving, prompted deliberation, or scaffolded performance.

Emergence is a second capability-attribution problem. Wei et al. (2022) define emergent abilities operationally as abilities that appear in larger models but not smaller ones. That framing explains why capability language is tempting: some scaling curves look like qualitative change. Schaeffer et al. (2023) show why the inference needs care. If a nonlinear or discontinuous metric makes gradual performance change look sudden, then the curve supports a claim about the interaction among model scale, task, and metric before it supports a claim about a qualitatively new capability.

The constructive lesson is to report the metric and task structure that make an emergence claim projectible. A sharp threshold under one scoring rule can license a claim about performance under that rule. It licenses a broader capability claim only when neighbouring metrics, item populations, and model families support the same interpretation.

Benchmark infrastructure can either sharpen or blur the attribution. MMLU uses many academic and professional subjects to measure breadth of multitask understanding, but it also shows lopsided performance and calibration problems (Hendrycks et al., 2020). HELM responds to narrow leaderboard culture by combining scenarios, metrics, and standardization, making visible that accuracy is only one desideratum among calibration, robustness, fairness, bias, toxicity, and efficiency (Liang et al., 2023). Dynabench pushes the same lesson dynamically: model ability should be probed through human-and-model-in-the-loop examples that expose failures after static benchmarks saturate (Kiela et al., 2021).

Agency is the system-boundary case. Agentic LLM surveys define the target as a system that receives natural-language input, reasons or plans, acts autonomously, uses tools or environments, and pursues goals (Plaat et al., 2025). The attribution therefore attaches to a model-system configuration rather than to a bare language model. Tool access, memory, planning horizon, feedback, permissions, and environment design help determine whether the system counts as agentic for a given evaluation.

The same base model can be part of different agency profiles. In an autocomplete interface, it predicts text without acting in the world. In a tool-using assistant, it can call APIs, revise plans, and affect an external state. In a multi-agent simulation, its behaviour depends on other agents and feedback from the environment. The agency attribution travels only with the relevant system boundary.

Classifying these terms as capability attributions turns the question "Does the model reason?" into a measurement question: which capability claim, licensed by which operationalization, projects to which domain, under which contamination and scaffold conditions? The same form applies to emergence and agency. Which threshold, under which metric, projects across which model families? Which model-system configuration, with which tools and permissions, supports which action claim? The constructive payoff is calibrated attribution: a capability term becomes useful when it predicts transfer, robustness, and failure boundaries better than the benchmark score alone.

8 Alignment and trustworthiness as umbrella governance constructs

Umbrella governance constructs aggregate lower-level constructs for governance purposes. Their role is closer to labels such as health or fitness than to a single behavioural kind. The label signals that a set of lower-level evaluations is being managed together for a stated institutional use, such as release, procurement, audit, or risk management. NIST's AI Risk Management Framework treats trustworthy AI in just this way, bundling validity and reliability, safety, security and resilience, accountability and transparency, explainability and interpretability, privacy, and fairness with harmful bias managed (National Institute of Standards and Technology, 2023).

The projectible claim is coordinative rather than behavioural. A coordinative claim concerns how lower-level evaluations are bundled for a decision, not one stable output pattern across tasks. A trustworthiness label predicts that a specified set of lower-level evaluations has been assembled for a decision, that the same bundle can be inspected again in a similar use, and that omitting one component changes the governance claim. A release decision and a procurement decision can therefore use the same umbrella word while licensing different projections.

Alignment is similarly target-dependent. One target is user preference: RLHF trains models to follow instructions by optimizing against preference data and reward models (Ouyang et al., 2022). Another target is truthfulness. A third is policy compliance. A fourth is longer-term user welfare or institutional risk management. These targets can reinforce one another, but the construct claim changes when the target changes.

Consider a deployed assistant that gives agreeable answers to user beliefs, corrects false factual claims, and refuses prohibited requests under a content policy. A preference evaluation, a truthfulness evaluation, and a policy-compliance evaluation can assign different scores to the same system. Calling the system aligned says little until the lower-level target is stated.

Sycophancy is the pressure case. A behaviour can satisfy one alignment target, such as immediate user preference, while failing another, such as truthfulness. Reward-model overoptimization gives the general proxy problem: optimizing a learned reward can outrun the target it was meant to stand in for (Gao et al., 2023). Sycophancy research makes the conflict concrete: preference data and preference models can reward responses that match user beliefs over truthful corrections (Sharma et al., 2024), and recent formal work argues that preference-based post-training can amplify this behaviour when the learned reward favours agreement (Shapira et al., 2026).

The pressure case matters because it can reverse the evaluative verdict without changing the surface behaviour. Agreement with a false user belief can be a success relative to a learned preference signal and a failure relative to truthfulness, calibration, or epistemic support. The choice of lower-level target determines which inference is licensed. The umbrella term shouldn't hide that target choice.

The question "Is sycophancy misalignment?" is therefore indexed to the lower-level construct in view. Relative to a narrow preference target, some sycophantic behaviour can be a local alignment success. Relative to truthfulness, calibration, user autonomy, or epistemic support, it's a failure. That's the point of treating alignment as an umbrella construct: it coordinates lower-level targets that can come apart.

Umbrella constructs are projectible in a coordinative sense. They predict which lower-level constructs should be evaluated together for a stated governance decision. A trustworthiness claim should license expectations about the bundled management of validity, reliability, safety, security, accountability, transparency, privacy, and fairness under a specified use. An alignment claim should say which lower-level target is in view and block projection to incompatible targets. Without that specification, the umbrella label makes a decision sound measured while leaving the measured construct underdescribed.

9 Conclusion

AI evaluation vocabulary has several inferential roles. Some terms mark output-centred failure profiles, some mark adversarial relations, some support capability attributions, and some coordinate governance constructs. Each type licenses a different inference, travels under different conditions, and fails in a characteristic way when evaluators read it as another type.

The typology's practical use is diagnostic. It asks what observation produced a label, what operationalization gave the label content, what inference the label licenses, and where that inference stops. Evaluation design, benchmark reporting, model comparison, and governance language should make those answers explicit. Who uses a label, and what decision it licenses, is part of the measurement system.

The typology is a starting point rather than a closed map. Bias is the obvious extension. It looks partly like an output profile, partly like a deployment-sensitive relation among model, population, task, and affected group, and partly like a governance construct. That complexity is exactly why the diagnostic matters. The useful question is what the term licenses in a given evaluation, what structure would let the inference travel, and what overgeneralization the label should block.

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References

Bean, A. M., Kearns, R. O., Romanou, A., Hafner, F. S., Mayne, H., Batzner, J., Foroutan, N., Schmitz, C., Korgul, K., Batra, H., Deb, O., Beharry, E., Emde, C., Foster, T., Gausen, A., Grandury, M., Han, S., Hofmann, V., Ibrahim, L., ... Mahdi, A. (2025). Measuring what matters: Construct validity in large language model benchmarks [NeurIPS 2025 Track on Datasets and Benchmarks]. *arXiv preprint arXiv:2511.04703*. https://doi.org/10.48550/arXiv.2511.04703

Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, *61*, 127–148. https://doi.org/10.1007/BF00385837

Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. https://doi.org/10.7551/mitpress/6396.003.0012

Chouldechova, A., Atalla, C., Barocas, S., Cooper, A. F., Corvi, E., Dow, P. A., Garcia-Gathright, J., Pangakis, N., Reed, S., Sheng, E., Vann, D., Vogel, M., Washington, H., & Wallach, H. (2024). A shared standard for valid measurement of generative AI systems' capabilities, risks, and impacts [NeurIPS 2024 Workshop on Statistical Foundations of LLMs and Foundation Models]. *arXiv preprint arXiv:2412.01934*. https://doi.org/10.48550/arXiv.2412.01934

Freiesleben, T. (2026). Establishing construct validity in LLM capability benchmarks requires nomological networks. *arXiv preprint arXiv:2603.15121*. https://doi.org/10.48550/arXiv.2603.15121

Gao, L., Schulman, J., & Hilton, J. (2023). Scaling laws for reward model overoptimization. *Proceedings of the 40th International Conference on Machine Learning*, *202*, 10835–10866. https://proceedings.mlr.press/v202/gao23h.html

Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. *arXiv preprint arXiv:2302.12173*. https://doi.org/10.48550/arXiv.2302.12173

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2020). Measuring massive multitask language understanding [ICLR 2021 conference paper]. *arXiv preprint arXiv:2009.03300*. https://doi.org/10.48550/arXiv.2009.03300

Huang, J., & Chang, K. C.-C. (2023). Towards reasoning in large language models: A survey. *Findings of the Association for Computational Linguistics: ACL 2023*, 1049–1065. https://doi.org/10.18653/v1/2023.findings-acl.67

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., & Liu, T. (2025). A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, *43*(2), 1–55. https://doi.org/10.1145/3703155

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, *55*(12), Article 248, 1–38. https://doi.org/10.1145/3571730

Kane, M. T. (2013). Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, *50*(1), 1–73. https://doi.org/10.1111/jedm.12000

Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, *195*(4), 1379–1396. https://doi.org/10.1007/s11229-015-0841-y

Kiela, D., Bartolo, M., Nie, Y., Kaushik, D., Geiger, A., Wu, Z., Vidgen, B., Prasad, G., Singh, A., Ringshia, P., Ma, Z., Thrush, T., Riedel, S., Waseem, Z., Stenetorp, P., Jia, R., Bansal, M., Potts, C., & Williams, A. (2021). Dynabench: Rethinking benchmarking in NLP. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4110–4124. https://doi.org/10.18653/v1/2021.naacl-main.324

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., Newman, B., Yuan, B., Yan, B., Zhang, C., Cosgrove, C., Manning, C. D., Ré, C., Acosta-Navas, D., Hudson, D. A., ... Koreeda, Y. (2023). Holistic evaluation of language models. *Transactions on Machine Learning Research*. https://openreview.net/forum?id=iO4LZibEqW

Manakul, P., Liusie, A., & Gales, M. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 9004–9017. https://doi.org/10.18653/v1/2023.emnlp-main.557

Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-t., Koh, P., Iyyer, M., Zettlemoyer, L., & Hajishirzi, H. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 12076–12100. https://doi.org/10.18653/v1/2023.emnlp-main.741

Mirzadeh, I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S., & Farajtabar, M. (2025). GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models [ICLR 2025 conference paper]. *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=AjXkRZIvjB

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST Artificial Intelligence Report No. NIST AI 100-1). National Institute of Standards and Technology. Gaithersburg, MD. https://doi.org/10.6028/NIST.AI.100-1

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P. F., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, *35*. https://proceedings.neurips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract.html

Plaat, A., van Duijn, M., van Stein, N., Preuss, M., van der Putten, P., & Batenburg, K. J. (2025). Agentic large language models, a survey [74 pages]. *Journal of Artificial Intelligence Research*, *84*, Article 29. https://doi.org/10.1613/jair.1.18675

Raji, I. D., Bender, E. M., Paullada, A., Denton, E., & Hanna, A. (2021). AI and the everything in the whole wide world benchmark. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, *1*. https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/084b6fbb10729ed4da8c3d3f5a3ae7c9-Abstract-round2.html

Schaeffer, R., Miranda, B., & Koyejo, S. (2023). Are emergent abilities of large language models a mirage? *Advances in Neural Information Processing Systems*, *36*. https://proceedings.

neurips.cc/paper_files/paper/2023/hash/adc98a266f45005c403b8311ca7e8bd7-Abstract-Conference.html

Shapira, I., Benade, G., & Procaccia, A. D. (2026). How RLHF amplifies sycophancy. *arXiv preprint arXiv:2602.01002*. https://doi.org/10.48550/arXiv.2602.01002

Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., Bowman, S., Durmus, E., Hatfield-Dodds, Z., Johnston, S., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., & Perez, E. (2024). Towards understanding sycophancy in language models. *International Conference on Learning Representations*. https://proceedings.iclr.cc/paper_files/paper/2024/hash/0105f7972202c1d4fb817da9f21a9663-Abstract-Conference.html

Shen, X., Chen, Z., Backes, M., Shen, Y., & Zhang, Y. (2024). Do Anything Now: Characterizing and evaluating in-the-wild jailbreak prompts on large language models. *arXiv preprint arXiv:2308.03825*. https://doi.org/10.48550/arXiv.2308.03825

Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021). Retrieval augmentation reduces hallucination in conversation. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 3784–3803. https://doi.org/10.18653/v1/2021.findings-emnlp.320

Vassilev, A., Oprea, A., Fordyce, A., & Anderson, H. (2024). *Adversarial machine learning: A taxonomy and terminology of attacks and mitigations* (NIST Artificial Intelligence Report No. NIST AI 100-2e2023). National Institute of Standards and Technology. Gaithersburg, MD. https://doi.org/10.6028/NIST.AI.100-2e2023

Wei, J., Tay, Y., Bommasani, R., Raffel, C., Zoph, B., Borgeaud, S., Yogatama, D., Bosma, M., Zhou, D., Metzler, D., Chi, E. H., Hashimoto, T., Vinyals, O., Liang, P., Dean, J., & Fedus, W. (2022). Emergent abilities of large language models. *Transactions on Machine Learning Research*. https://openreview.net/forum?id=yzkSU5zdwD

Xiao, Y., & Wang, W. Y. (2021). On hallucination and predictive uncertainty in conditional language generation. *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, 2734–2744. https://doi.org/10.18653/v1/2021.eacl-main.236

Yi, S., Liu, Y., Sun, Z., Cong, T., He, X., Song, J., Xu, K., & Li, Q. (2024). Jailbreak attacks and defenses against large language models: A survey. *arXiv preprint arXiv:2407.04295*. https://doi.org/10.48550/arXiv.2407.04295

Zeng, Y., Lin, H., Zhang, J., Yang, D., Jia, R., & Shi, W. (2024). How johnny can persuade LLMs to jailbreak them: Rethinking persuasion to challenge AI safety by humanizing LLMs. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 14322–14350. https://doi.org/10.18653/v1/2024.acl-long.773

Zou, A., Wang, Z., Carlini, N., Nasr, M., Kolter, J. Z., & Fredrikson, M. (2023). Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*. https://doi.org/10.48550/arXiv.2307.15043